ADV-DEVOPS EXPERIMENT-6

Aim : To understand terraform lifecycle, core concepts/terminologies, and install it on a windows machine.

Theory :

Terraform Overview:

Terraform is an open-source IaC tool developed by **HashiCorp**. It allows users to define both cloud and on-prem resources using a declarative configuration language called **HashiCorp Configuration Language (HCL)**. Terraform supports multiple cloud providers, including AWS, Azure, and GCP.

Using Terraform, users can:

- **Build** infrastructure (e.g., EC2 instances, S3 buckets)
- **Change** or update existing infrastructure
- **Destroy** infrastructure when it is no longer needed

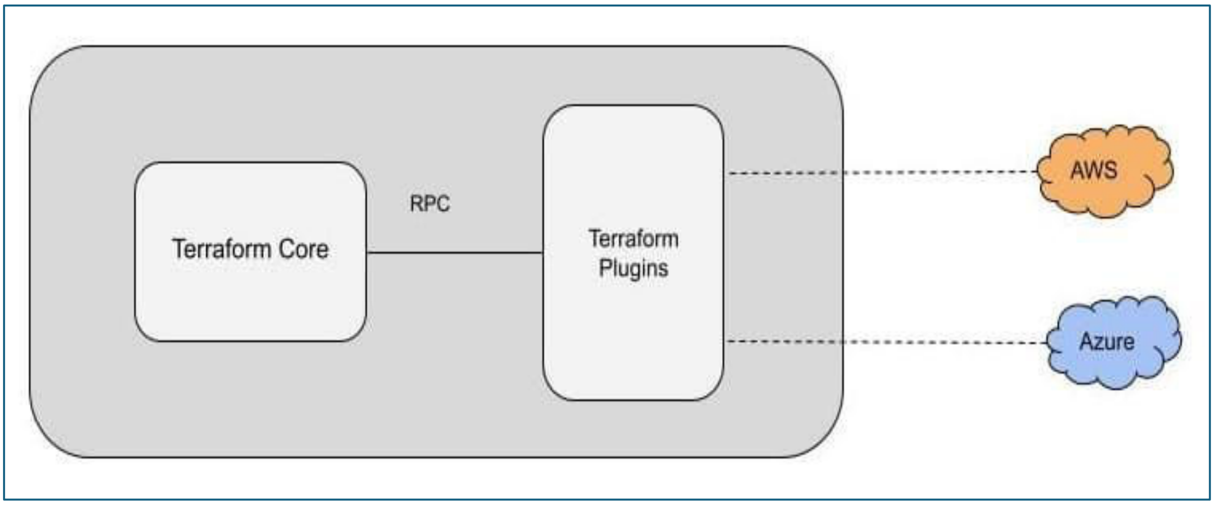
Terraform Core Concepts:

- **Providers**: These are plugins used to interact with APIs of different services (e.g., AWS, Azure). For this experiment, we use the AWS provider.
- **Resources**: Basic building blocks in Terraform representing infrastructure components (like EC2 instances, VPCs, etc.).
- **Variables**: Input parameters that make Terraform configurations reusable.
- **State**: Terraform maintains a state file (terraform.tfstate) to keep track of the current infrastructure.
- **Plan & Apply**:
  - **terraform plan:** Shows what changes Terraform will make without applying them.
  - **terraform apply:** Applies the planned changes to build or modify infrastructure.
- **Destroy**:
  - **terraform destroy:** Removes all infrastructure components defined in the Terraform configuration.

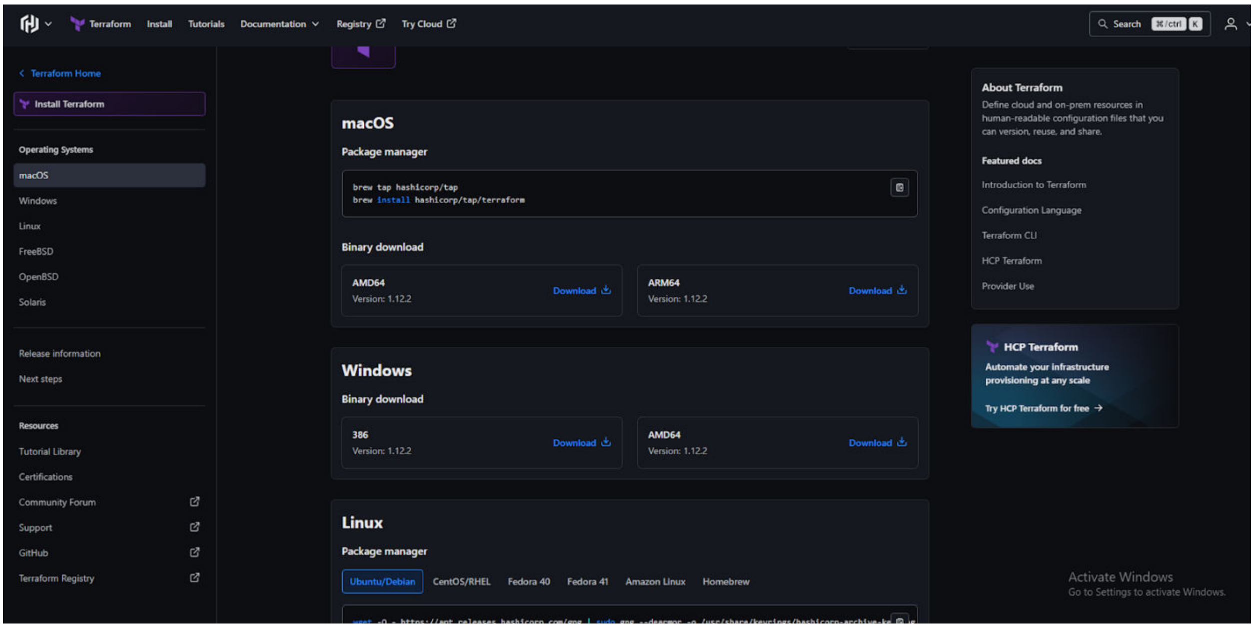
Lifecycle of Terraform Usage:

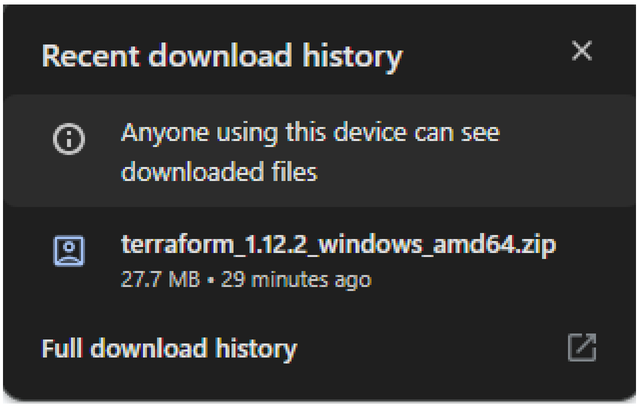
The standard lifecycle in this experiment involves three main phases:

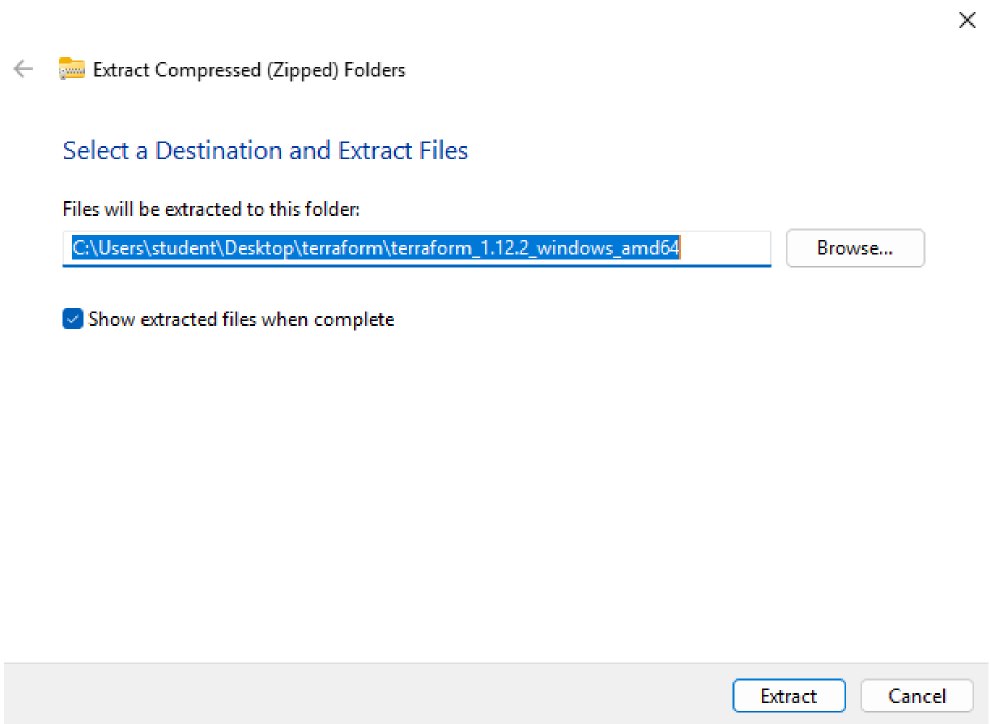
- **Build**: Write Terraform configuration and apply it to provision AWS resources.
- **Change**: Modify the configuration to update resources, then apply again to reflect changes.
- **Destroy**: Tear down the infrastructure using terraform destroy when it’s no longer needed.

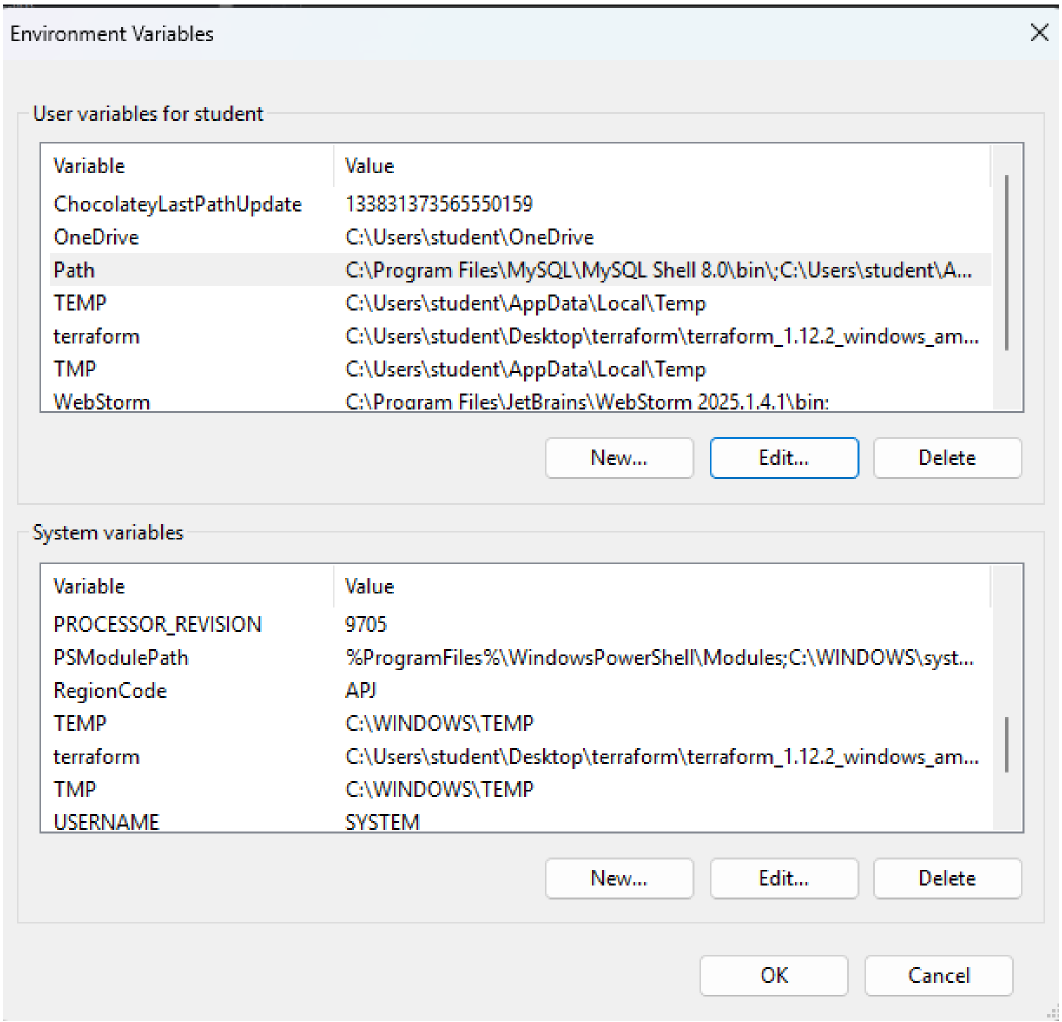


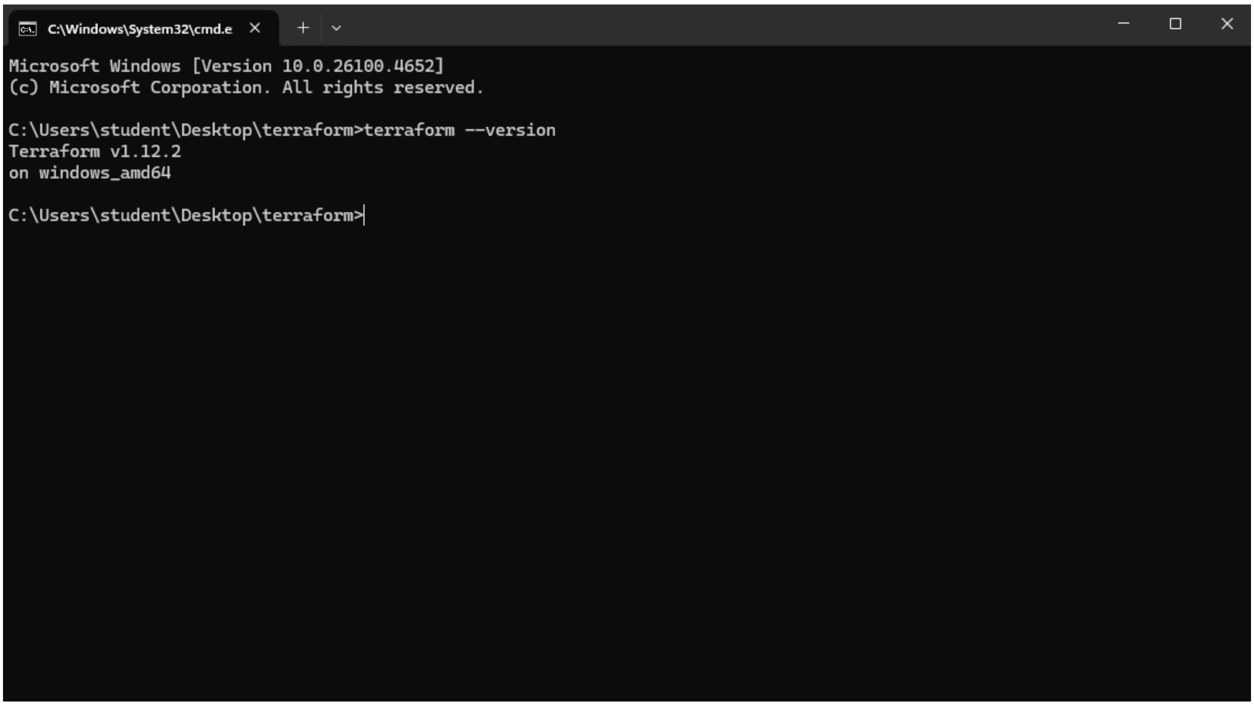
<u>Procedure :</u>






Extract Compressed (Zipped) Folders
Select a Destination and Extract Files
Files will be extracted to this folder:
C:\Users\student\Desktop\terraform\terraform_1.12.2_windows_amd64
Browse...
Show extracted files when complete
Extract
Cancel


Environment Variables
User variables for student
Variable
Value
ChocolateyLastPathUpdate
133831373565550159
OneDrive
C:\Users\student\OneDrive
Path
C:\Program Files\MySQL\MySQL Shell 8.0\bin\;C:\Users\student\A...
TEMP
C:\Users\student\AppData\Local\Temp
terraform
C:\Users\student\Desktop\terraform\terraform_1.12.2_windows_am...
TMP
C:\Users\student\AppData\Local\Temp
WebStorm
C:\Program Files\JetBrains\WebStorm 2025.1.4.1\bin;
New...
Edit...
Delete
System variables
Variable
Value
PROCESSOR_REVISION
9705
PSModulePath
%ProgramFiles%\WindowsPowerShell\Modules;C:\WINDOWS\syst...
RegionCode
APJ
TEMP
C:\WINDOWS\TEMP
terraform
C:\Users\student\Desktop\terraform\terraform_1.12.2_windows_am...
TMP
C:\WINDOWS\TEMP
USERNAME
SYSTEM
New...
Edit...
Delete
OK
Cancel



Conclusion :

Terraform enables developers and operations teams to define the desired state of their infrastructure, allowing Terraform to manage the entire lifecycle of resources like virtual machines, storage, and networking across various platforms such as AWS, Azure, and Google Cloud.